Chapter 14: Introduction to Inference

The Practice of Statistics in the Life Sciences
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Part III: Foundations of Inference

From Sampling Distributions to Statistical Inference

Chapter 13 showed that a sample statistic varies from one random sample to another.

This sampling variation means that a statistic such as $\bar{x}$ or $\hat{p}$ will not usually equal the corresponding population parameter exactly.

However, the sampling distribution describes how the statistic is expected to vary. We can use this information to measure the uncertainty associated with a sample result.

Central Question

What can sample data tell us about an unknown population parameter?

Statistical inference uses sample data to draw conclusions about a population while accounting for random sampling variation.

Two Main Goals of Statistical Inference

Statistical inference has two closely related goals.

Estimation

A **confidence interval** uses sample data to estimate an unknown population parameter. It provides a range of plausible values and communicates the precision of the estimate.

Testing a Claim

A **hypothesis test** evaluates how compatible the observed sample data are with a specific claim about a population parameter. The evidence is assessed by comparing the observed statistic with the sampling distribution expected under that claim.

Both methods use sampling distributions to account for the variation that occurs from one random sample to another.

Why Begin with a Known Population Standard Deviation?

When making inferences about a population mean, the population standard deviation σ is usually unknown.

In this chapter, we initially assume that σ is known. Although this situation is uncommon in practice, the assumption allows us to focus on the main ideas of statistical inference.

- How sampling distributions quantify uncertainty
- How confidence intervals are constructed and interpreted
- How evidence against a population claim is evaluated
- How sample size and confidence level affect precision

When σ is unknown, it must be estimated using the sample standard deviation s. This leads to methods based on the t distribution, which will be introduced later.

Point Estimates and Interval Estimates

A sample statistic can be used to estimate an unknown population parameter.

For a population mean μ, the natural point estimate is the sample mean

$$\bar{x}$$

- A **point estimate** gives a single numerical estimate of the parameter
- An **interval estimate** gives a range of plausible values for the parameter

A confidence interval has the general form

$$\text{point estimate} \pm \text{margin of error}$$

The margin of error describes the uncertainty produced by random sampling variation.

Example 14.1: Estimating a Population Mean

Question

A hospital's pediatric ward survey selects a random sample of $n = 217$ eight year old patients from the population of interest. The sample mean height is $\bar{x} = 132.5$ cm. For this introductory example, assume that the population standard deviation is known to be $\sigma = 10$ cm. How can the sample be used to estimate the unknown population mean height μ while accounting for random sampling variation?

The sample mean provides the point estimate

$$\bar{x} = 132.5 \text{ cm}$$

However, the point estimate alone does not describe how much $\bar{x}$ would vary across repeated random samples.

Example 14.1: Calculating the Standard Error

For random samples of size n, the sampling distribution of $\bar{x}$ has centre and standard deviation

$$\mu_{\bar{x}} = \mu \quad \text{and} \quad \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

This standard deviation describes the typical distance between a sample mean and the population mean. In this setting, it is called the **standard error of the sample mean**.

For the height data

$$\sigma_{\bar{x}} = \frac{10}{\sqrt{217}} \approx 0.679 \text{ cm}$$

Therefore, sample means from repeated random samples of 217 patients would typically vary by about 0.679 cm around μ.

Example 14.2: Finding an Approximate Margin of Error

Because the sample size is large, the sampling distribution of $\bar{x}$ is approximately Normal. The 68–95–99.7 rule states that approximately 95% of a Normal distribution lies within two standard deviations of its mean.

Therefore, approximately 95% of sample means will fall within two standard errors of the population mean.

$$\bar{x} \text{ is usually within } 2\sigma_{\bar{x}} \text{ of } \mu$$

The approximate margin of error is

$$m = 2\sigma_{\bar{x}} = 2(0.679) \approx 1.36 \text{ cm}$$

The value 2 is being used as an approximate 95% critical value. The more precise value is 1.96.

Example 14.2: Constructing the Confidence Interval

The approximate 95% confidence interval has the form

$$\bar{x} \pm m$$

Substituting the sample mean and margin of error gives

$$132.5 \pm 1.36$$

The lower endpoint is

$$132.5 - 1.36 = 131.14$$

The upper endpoint is

$$132.5 + 1.36 = 133.86$$

After rounding appropriately, the approximate 95% confidence interval is

$$\boxed{(131.1,\ 133.9)\ \text{cm}}$$

Example 14.2: Interpreting the Confidence Interval

Interpretation

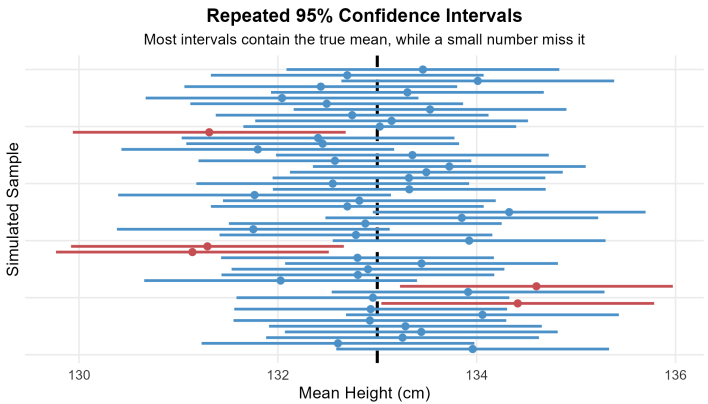
Using this confidence interval procedure, we are approximately 95% confident that the population mean height of eight year old patients in the population of interest is between 131.1 cm and 133.9 cm.

The confidence level describes the reliability of the **procedure** used to construct the interval.

- The population mean μ is fixed but unknown
- The sample mean and confidence interval vary from sample to sample
- Some intervals contain μ, while others do not

The interval does not state that 95% of individual heights lie between 131.1 cm and 133.9 cm. It is an interval for the population mean, not for individual observations.

Example 14.2: Interpreting the Confidence Interval



Each interval is constructed from a different random sample.

- The vertical reference line represents the fixed population mean
- The intervals change because the sample means change
- Most intervals contain the population mean
- A small proportion fail to contain it

What Does 95% Coverage Mean?

A 95% confidence procedure has a **long-run coverage rate of approximately** 95% when its assumptions are satisfied.

If the sampling process were repeated many times and a confidence interval were constructed from each sample, approximately 95% of those intervals would contain the true population parameter.

What It Does Not Mean

After one particular interval has been calculated, it is not correct to say that there is a 95% probability that the fixed parameter lies inside that interval.

Anatomy of a Confidence Interval

Most confidence intervals have the form

$$\text{estimate} \pm \text{margin of error}$$

The margin of error can be separated into two components

$$\text{margin of error} = \text{critical value} \times \text{standard error}$$

Therefore, the general structure is

$$\boxed{\text{confidence interval} = \text{estimate} \pm (\text{critical value})(\text{standard error})}$$

Three ideas determine the interval:

- The **estimate** determines the centre
- The **standard error** describes sampling variability
- The **confidence level** determines the required critical value

The Estimate: Locating the Centre

When estimating a population mean μ, the natural point estimate is the sample mean

$$\bar{x}$$

Therefore, a confidence interval for μ is centred at $\bar{x}$.

Under the random-sampling model, the sample mean is an **unbiased estimator** of the population mean

$$E(\bar{X}) = \mu$$

Unbiasedness means that the sample mean is centred at μ across repeated random samples. It does not guarantee that the mean from one particular sample will be close to μ.

The Standard Error: Measuring Sampling Variability

When the population standard deviation σ is known, the standard error of the sample mean is

$$\text{SE}(\bar{x}) = \frac{\sigma}{\sqrt{n}}$$

The standard error describes how much the sample mean would typically vary from one random sample to another.

- A smaller standard error means that $\bar{x}$ is usually closer to μ
- A larger standard error means that sample means vary more across samples
- Larger random samples produce smaller standard errors

The Confidence Level and Critical Value

The **confidence level** is the intended long-run coverage rate of the confidence interval procedure. For a confidence level C, the critical value z^* is chosen so that the central area under the Standard Normal curve equals C:

$$P(-z^* < Z < z^*) = C$$

If $\alpha = 1 - C$ then the area in each tail is $\frac{\alpha}{2}$

Confidence level	**Area in each tail**	**Critical value z^***
90%	0.050	1.645
95%	0.025	1.960
99%	0.005	2.576

A higher confidence level requires a larger critical value.

The Margin of Error

For a confidence interval for a population mean with known σ, the margin of error is

$$m = z^* \, \mathrm{SE}(\bar{x}) = z^* \frac{\sigma}{\sqrt{n}}$$

The resulting confidence interval is

$$\boxed{\bar{x} \pm z^* \frac{\sigma}{\sqrt{n}}}$$

Precision, Confidence and Interval Width

The formula shows how each component affects precision.

- Increasing the sample size n reduces the standard error and produces a narrower interval
- Increasing the confidence level increases z^* and produces a wider interval
- Greater population variability σ produces a wider interval

The Confidence–Precision Trade-off

For the same data, greater confidence requires a wider interval. A narrower interval provides greater precision but, without increasing the sample size, it must use a lower confidence level.

Because the margin of error depends on $\sqrt{n}$, multiplying the sample size by 4 reduces the margin of error by one-half.

How to Interpret a Confidence Interval

Suitable Interpretation

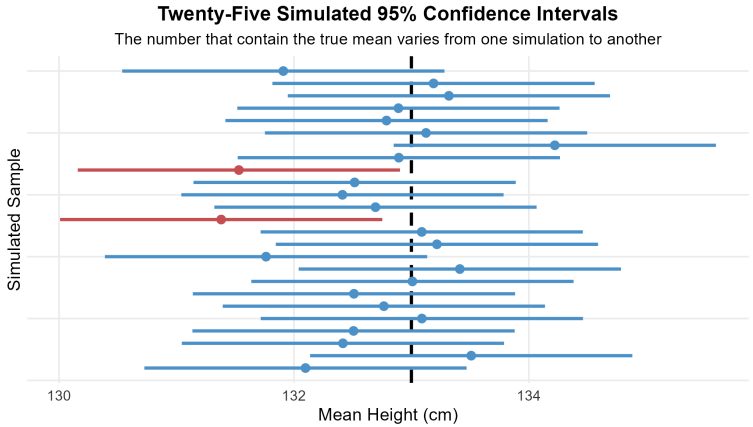
We are 95% confident that the unknown population parameter lies between the reported endpoints. The confidence comes from using a procedure that captures the true parameter in approximately 95% of repeated random samples when the required conditions are satisfied.

Common Misinterpretations

A 95% confidence interval does not mean that:

- There is a 95% probability that the fixed parameter lies in this particular interval
- 95% of individual observations lie between the endpoints
- 95% of future observations will lie between the endpoints

Twenty-Five Simulated 95% Confidence Intervals



Each horizontal interval was constructed from a different random sample.

- The population parameter remains fixed
- The interval changes from sample to sample
- An interval contains the parameter when it crosses the reference line
- An interval misses the parameter when it does not cross the reference line

The Four-Step Process for Confidence Intervals

A consistent process keeps the statistical calculations connected to the scientific question.

1. **STATE**
   Identify the population, the parameter being estimated and the desired confidence level.
2. **PLAN**
   Select an appropriate confidence-interval procedure and assess whether its conditions are reasonably satisfied.
3. **SOLVE**
   Calculate the estimate, standard error, critical value, margin of error and confidence interval.
4. **CONCLUDE**
   Interpret the interval in the context of the population and acknowledge any important limitations.

Conditions for a Confidence Interval for μ

Fo one sample z confidence interval we have these conditions.

Conditions

- **Random Sampling**: The data should come from a random sample or another sampling process that reasonably represents the population of interest.
- **Independence**: The observations should be independent. When sampling without replacement, the sample size should generally be no more than 10% of the population $n \leq 0.10N$.
- **Approximately Normal Sampling Distribution**: The population should be approximately Normal, or the sample size should be sufficiently large for the Central Limit Theorem to apply. Strong skewness and extreme outliers require additional caution.
- **Known Standard Devation**: For the z procedure (introduced here), the population standard deviation σ is also treated as known.

Example 14.3: Leaf Surface Area

Question

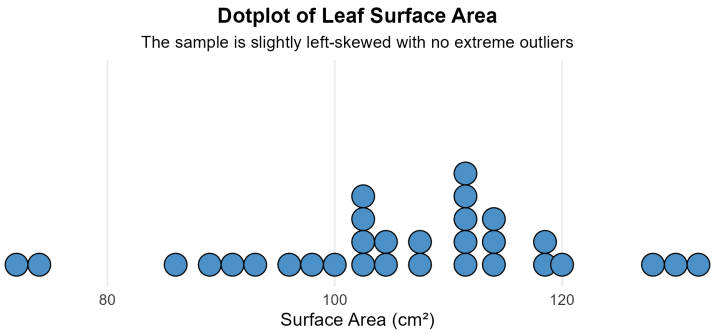
A botanist wants to estimate the mean surface area of mature common bean leaves in a greenhouse containing more than 310 mature leaves. A simple random sample of 31 leaves is selected. The population standard deviation is treated as known at $\sigma = 15$ cm^2. What is a 95% confidence interval for the population mean leaf surface area?

114	100	104	89	102	91	114	114
103	105	108	130	120	132	111	128
118	119	86	72	111	103	74	112
107	103	98	96	112	112	93	

STATE

The parameter μ is the mean surface area, in cm^2, of all mature common bean leaves in this greenhouse population.

Example 14.3: Examining the Sample Data



Consider:

- The overall shape
- The amount of skewness
- Possible gaps or clusters
- Any extreme outliers

The data are somewhat left-skewed and no clear extreme outlier.

Example 14.3: Planning and Checking Conditions

PLAN

A one sample z confidence interval for μ will be used because the population standard deviation is treated as known.

- **Random sampling:** The 31 leaves were selected using a simple random sample
- **Independence:** The greenhouse contains more than 310 mature leaves, so the sample contains less than 10% of the population $31 \leq 0.10N$
- **Shape and sample size:** The sample distribution is somewhat left-skewed but has no extreme outliers. With $n = 31$, the Central Limit Theorem provides reasonable support for an approximately Normal sampling distribution of $\bar{x}$
- **Known population standard deviation:**

$$\sigma = 15 \text{ cm}^2$$

Example 14.3: Calculating the Interval

SOLVE

The sample mean is

$$\bar{x} = \frac{3281}{31} \approx 105.84 \text{ cm}^2$$

The standard error is

$$\text{SE}(\bar{x}) = \frac{\sigma}{\sqrt{n}} = \frac{15}{\sqrt{31}} \approx 2.694 \text{ cm}^2$$

For a 95% confidence interval

$$z^* = 1.960$$

The margin of error is

$$m = 1.960(2.694) \approx 5.28 \text{ cm}^2$$

Example 14.3: Interval and Conclusion

The confidence interval is

$$\bar{x} \pm m$$

$$105.84 \pm 5.28$$

Therefore

$$(100.56,\ 111.12)\ \text{cm}^2$$

CONCLUDE

We are 95% confident that the mean surface area of all mature common bean leaves in this greenhouse population is between approximately 100.6 cm^2 and 111.1 cm^2.

Example 14.3: Confidence Interval in R

```
# Observed leaf surface areas in square centimetres
leaf_area <- c(114, 100, 104, 89, 102, 91, 114, 114, 103, 105, 108, 130, 120, 132, 111, 128,
  118, 119, 86, 72, 111, 103, 74, 112, 107, 103, 98, 96, 112, 112, 93
)

# Population standard deviation
sigma <- 15

# Sample information, critical value and margin of error
x_bar <- mean(leaf_area)
n <- length(leaf_area)
standard_error <- sigma / sqrt(n)
z_star <- qnorm(0.975)
margin_error <- z_star * standard_error

# Calculate and display the confidence interval
confidence_interval <- c(lower = x_bar - margin_error, upper = x_bar + margin_error)
round(confidence_interval, 2)

> lower  upper
> 100.56 111.12
```

Looking Beyond the Simple Conditions

In most real studies, the population standard deviation is unknown.

Later chapters replace σ with the sample standard deviation s and use Student's t distribution.

The central logic remains the same: an estimate is combined with a standard error and an appropriate critical value.

Break Time



From Estimation to Hypothesis Testing

A confidence interval begins with an unknown population parameter and asks which values are plausible based on the sample data.

A hypothesis test begins with a specific claim about a population parameter and asks whether the observed data are compatible with that claim.

Central Question

If the population claim were true, how unusual would the observed sample result be?

A result that would be very unusual under the stated claim provides evidence against that claim. A result that is not unusual does not provide strong evidence against it.

The Null and Alternative Hypotheses

A hypothesis test compares two competing statements about a population parameter.

Null Hypothesis

The **null hypothesis**, denoted by H_0, gives the specific population value or model being evaluated.

$$H_0 : \mu = \mu_0$$

Alternative Hypothesis

The **alternative hypothesis**, denoted by H_a, describes the competing population claim for which evidence is being sought.

The test begins by temporarily assuming that H_0 is true. The observed data are then compared with what would be expected under this assumption.

One-Sided and Two-Sided Alternatives

The form of the alternative hypothesis depends on the scientific question.

Types of Hypothesis Tests

- **Lower-Tail Test**: A decrease or value below the reference level is being investigated.

$$H_a : \mu < \mu_0$$

- **Upper-Tail Test**: An increase or value above the reference level is being investigated.

$$H_a : \mu > \mu_0$$

- **Two-Sided Test**: A difference in either direction is being investigated.

$$H_a : \mu \neq \mu_0$$

Type I and Type II Errors

A hypothesis test uses sample data, so an incorrect decision is possible.

	H_0 **is true**	H_0 **is false**
Reject H_0	Type I error	Correct decision
Fail to reject H_0	Correct decision	Type II error

- A **Type I error** occurs when a true null hypothesis is rejected. Its probability is denoted by α
- A **Type II error** occurs when a false null hypothesis is not rejected. Its probability is denoted by β

The consequences of the two errors depend on the context. It is also important to say **fail to reject** H_0, rather than **accept** H_0, because the data may simply provide insufficient evidence against it.

Examples 14.4 and 14.5: Phosphorus Levels

Question

Blood phosphorus levels in the general adult population are modelled as Normal with mean 1.2 mmol/L and standard deviation 0.1 mmol/L. Researchers suspect that the population mean may be lower among adults aged 75 to 79. They randomly select 12 patient records from a large health system population in this age group and record one phosphorus measurement from each patient. Assume that the population standard deviation remains $\sigma = 0.1$ mmol/L. Does the sample provide evidence that the population mean phosphorus level for this older group is below 1.2 mmol/L?

1.26	1.39	1.00	1.00	1.00	1.10
0.87	1.23	1.19	1.29	1.03	1.18

Examples 14.4 and 14.5: Phosphorus Levels

The sample mean phosphorus level is

$$\bar{x} = \frac{13.54}{12} \approx 1.1283 \text{ mmol/L}$$

The observed difference from the reference value is

$$\bar{x} - \mu_0 = 1.1283 - 1.2 = -0.0717 \text{ mmol/L}$$

The sample mean is below 1.2 mmol/L. However, a difference between $\bar{x}$ and μ can occur because of random sampling variation.

Question for the Test

If the population mean were actually 1.2 mmol/L, how unusual would it be to obtain a sample mean of 1.1283 mmol/L or lower from a random sample of 12 patients?

Examples 14.4 and 14.5: Phosphorus Levels

Let μ represent the population mean phosphorus level among adults aged 75 to 79 in the health-system population.

The null hypothesis states that the population mean equals the general-adult reference value:

$$H_0 : \mu = 1.2 \text{ mmol/L}$$

The alternative hypothesis represents the researchers' suspicion that the population mean is lower:

$$H_a : \mu < 1.2 \text{ mmol/L}$$

This is a **lower-tail test** because only values below 1.2 mmol/L support the alternative hypothesis.

Example 14.6: Aspirin Tablet Manufacturing

Question

A pharmaceutical manufacturing process is designed to produce aspirin tablets with a population mean active ingredient amount of 325 mg. Underfilling may reduce the effectiveness of a tablet, while overfilling may increase the risk of adverse effects and violate manufacturing specifications. What hypotheses should be used to evaluate whether the population mean active-ingredient amount differs from the target?

Let μ represent the population mean active ingredient amount for all tablets produced by the manufacturing process.

Because deviations above or below the target are both relevant, the appropriate test must consider both directions.

Example 14.6: Tablet Manufacturing

The null hypothesis states that the manufacturing process is centred at the target amount:

$$H_0 : \mu = 325 \text{ mg}$$

The alternative hypothesis states that the population mean differs from the target:

$$H_a : \mu \neq 325 \text{ mg}$$

Interpretation

This is a **two-sided test** because both underfilling and overfilling are departures from the intended manufacturing target.

The hypotheses refer to the population mean μ, not to the sample mean $\bar{x}$. The sample mean will be used as evidence when evaluating the hypotheses.

The Logic of a Hypothesis Test

A hypothesis test evaluates the observed sample data under the temporary assumption that the null hypothesis is true.

The reasoning follows a consistent sequence:

1. Assume that H_0 is true
2. Determine the sampling distribution expected under H_0
3. Compare the observed statistic with this null distribution
4. Calculate the probability of obtaining a result at least as extreme as the observed result
5. Use this probability to evaluate the evidence against H_0

Central Idea

A result that would be unusual when H_0 is true provides evidence against H_0.

The Test Statistic

For a one-sample z test of a population mean with known σ, the test statistic is

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

Each part of the formula has a specific role:

- $\bar{x}$ is the observed sample mean
- μ_0 is the population mean stated by the null hypothesis
- $\bar{x} - \mu_0$ is the observed difference from the null value
- $\sigma/\sqrt{n}$ is the standard error expected under the null model

The resulting z statistic measures how many standard errors the observed sample mean is above or below the null value.

- A positive z indicates that $\bar{x} > \mu_0$
- A negative z indicates that $\bar{x} < \mu_0$
- A value far from 0 represents stronger disagreement with H_0

The P-Value

Definition

The **P-value** is the probability of obtaining a test statistic at least as extreme as the observed statistic in the direction or directions specified by the alternative hypothesis. This probability is calculated under the assumption that the null hypothesis and model conditions are true.

The phrase **at least as extreme** includes the observed result and results that would provide even stronger evidence against H_0.

The P-value is calculated from the null sampling distribution. It therefore answers the following question:

How unusual is the observed result if the null hypothesis is true?

The Alternative Hypothesis Determines the Tail

The direction of the alternative hypothesis determines which area of the null distribution is used to calculate the P-value.

Alternative	**Type of test**	**P-value**
$H_a : \mu < \mu_0$	Lower-tail	$P(Z \leq z_{\text{obs}})$
$H_a : \mu > \mu_0$	Upper-tail	$P(Z \geq z_{\text{obs}})$
$H_a : \mu \neq \mu_0$	Two-sided	$P(\lvert Z \rvert \geq \lvert z_{\text{obs}} \rvert)$

For a symmetric Standard Normal distribution, the two-sided P-value can be calculated as

$$2P(Z \geq |z_{\text{obs}}|)$$

Interpreting P-Values

- A **small P-value** indicates that the observed result would be unusual if H_0 were true. It therefore provides stronger evidence against H_0
- A **large P-value** indicates that the observed result remains reasonably compatible with H_0
- A large P-value does not prove that H_0 is true. The study may have insufficient evidence to detect a meaningful departure from H_0

What a P-Value Is Not

A P-value is not

- The probability that H_0 is true
- The probability that H_a is true
- The probability that the observed result occurred only because of chance
- A measure of the practical, clinical or biological importance of an effect

Example 14.7: One-Sided Phosphorus Test

For the phosphorus study, let μ represent the population mean phosphorus level among adults aged 75 to 79. The hypotheses are

$$H_0 : \mu = 1.2 \text{ mmol/L}$$

$$H_a : \mu < 1.2 \text{ mmol/L}$$

The sample information is

$$n = 12 \qquad \bar{x} \approx 1.1283 \text{ mmol/L} \qquad \sigma = 0.1 \text{ mmol/L}$$

Question

Assuming that H_0 is true, what is the probability of obtaining a sample mean of 1.1283 mmol/L or lower from a random sample of 12 patients?

Example 14.7: The Null Sampling Distribution

Under the null hypothesis, the sampling distribution of $\bar{x}$ is centred at the null value

$$\mu_{\bar{x}} = \mu_0 = 1.2 \text{ mmol/L}$$

The standard error is

$$\text{SE}(\bar{x}) = \frac{\sigma}{\sqrt{n}} = \frac{0.1}{\sqrt{12}} \approx 0.0289 \text{ mmol/L}$$

Therefore, under H_0, sample means are expected to vary around 1.2 mmol/L with a standard error of approximately 0.0289 mmol/L.

The next step is to express the observed difference between $\bar{x}$ and μ_0 in standard error units.

Example 14.7: Calculating the Test Statistic

The observed difference from the null value is

$$\bar{x} - \mu_0 = 1.1283 - 1.2 = -0.0717 \text{ mmol/L}$$

The test statistic is

$$z_{\text{obs}} = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

$$z_{\text{obs}} = \frac{1.1283 - 1.2}{0.1/\sqrt{12}} \approx -2.48$$

The observed sample mean is approximately 2.48 standard errors below the mean specified by H_0. Because the alternative hypothesis is

$$H_a : \mu < 1.2$$

the P-value is the Standard Normal area to the left of $z = -2.48$.

Example 14.7: P-Value and Conclusion

The one-sided P-value is

$$P(Z \leq -2.48) = N(-2.48) = 1 - N(2.48) = 1 - 0.9934 = 0.0066$$

Interpretation of the P-Value

If the population mean phosphorus level were 1.2 mmol/L, the probability of obtaining a sample mean of approximately 1.1283 mmol/L or lower from a random sample of 12 patients would be about 0.0066 or 0.66%.

Conclusion

The data provide strong evidence against H_0 and support the conclusion that the population mean phosphorus level among adults aged 75 to 79 in the health-system population is below 1.2 mmol/L, provided that the model conditions are reasonably satisfied.

Example 14.7: One-Sided Test in R

```
# Observed phosphorus measurements in mmol/L
phosphorus <- c(1.26, 1.39, 1.00, 1.00, 1.00, 1.10, 0.87, 1.23, 1.19, 1.29, 1.03, 1.18)

# Null value and known population standard deviation
mu_0 <- 1.2
sigma <- 0.1

# Sample information
x_bar <- mean(phosphorus)
n <- length(phosphorus)
standard_error <- sigma / sqrt(n)

# Test statistic and p-value
z_observed <- (x_bar - mu_0) / standard_error
p_value <- pnorm(z_observed)

# Display the results
results <- c(sample_mean = x_bar, standard_error = standard_error,
  z_statistic = z_observed, p_value = p_value)
```

```
 sample_mean standard_error     z_statistic           p_value
      1.1283         0.0289         -2.4826            0.0065
```

Example 14.8: Two-Sided Aspirin Test

Question

A manufacturing process targets $\mu = 325$ mg with known $\sigma = 5$ mg. A random sample of $n = 10$ tablets has mean $\bar{x} = 326.9$ mg. The hypotheses are

$$H_0 : \mu = 325$$

$$H_a : \mu \neq 325$$

What is the two-sided P-value?

Example 14.8: Standard Error and Test Statistic

Under H_0, the sampling distribution of $\bar{x}$ is centred at 325 mg. Its standard error is

$$\mathrm{SE}(\bar{x}) = \frac{\sigma}{\sqrt{n}} = \frac{5}{\sqrt{10}} \approx 1.581 \text{ mg}$$

The test statistic is

$$z_{\text{obs}} = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{326.9 - 325}{1.581} \approx 1.202$$

The sample mean is approximately 1.202 standard errors above the target value. Because the alternative hypothesis is two-sided, equally extreme results above or below 325 mg must be included in the P-value.

Example 14.8: Standard Error and Test Statistic

For a two-sided test, the p-value includes both tails of the Standard Normal distribution:

$$P(|Z| \geq 1.202) = P(Z \leq -1.202) + P(Z \geq 1.202) = N(-1.202) + (1 - N(1.202)) \approx 0.2295$$

- If the population mean active ingredient amount were 325 mg, there would be approximately a 22.95% probability of obtaining a sample mean at least as far from 325 mg as 326.9 mg in either direction.
- Since

  $$0.2295 > 0.05$$

  the result is not statistically significant at the $\alpha = 0.05$ significance level. We fail to reject H_0.
- The sample does not provide sufficient evidence that the population mean active-ingredient amount differs from 325 mg. This conclusion does not prove that the process mean is exactly 325 mg.

Example 14.8: Standard Error and Test Statistic

```
# Null value and known population standard deviation
mu_0 <- 325
sigma <- 5

# Sample information
n <- 10
x_bar <- 326.9

# Standard error and test statistic
standard_error <- sigma / sqrt(n)
z_observed <- (x_bar - mu_0) / standard_error

# Two-sided P-value
# abs() measures distance from zero in either direction.
upper_tail <- 1 - pnorm(abs(z_observed))
p_value <- 2 * upper_tail

# Display the results
results <- c(standard_error = standard_error, z_statistic = z_observed, p_value = p_value)
round(results, 4)
standard_error     z_statistic          p_value
        1.5811          1.2017           0.2295
```

The Significance Level α

The **significance level**, denoted by α, is the decision threshold chosen before examining the sample results.

The p-value is compared with α:

- If $P \leq \alpha$, the result is **statistically significant** and H_0 is rejected
- If $P > \alpha$, the result is **not statistically significant** and we fail to reject H_0

Common significance levels include

$$\alpha = 0.05$$

and

$$\alpha = 0.01$$

The value of α should reflect the context and the consequences of making a Type I error. It should not be selected after seeing the p-value.

What Does α Mean?

For a correctly calibrated hypothesis test, α controls the long-run probability of rejecting H_0 when H_0 is true.

$$\alpha = P(\text{reject } H_0 \mid H_0 \text{ is true})$$

Therefore

$$\alpha = 0.05$$

means that, over many repetitions in which H_0 is true, approximately 5% of the tests would incorrectly reject it.

The One-Sample z Test for a Mean

The null hypothesis is

$$H_0 : \mu = \mu_0$$

The test statistic is

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

The statistic measures the observed difference from the null value in standard error units.

Conditions

- The data come from a random or reasonably representative sample
- The observations are independent
- The population is Normal or the sample is sufficiently large for the Central Limit Theorem
- The population standard deviation σ is treated as known

Example 14.9: Body Temperature

Question

An historical reference value for mean human body temperature is 98.6°F. A later study records one oral temperature from each of $n = 130$ healthy adult volunteers. The sample mean is

$$\bar{x} = 98.25°\text{F}$$

Assume that the observations are independent and reasonably representative of the population of interest. Also assume that the population standard deviation is known to be

$$\sigma = 0.6°\text{F}$$

At the $\alpha = 0.05$ significance level, do the data provide evidence that the population mean body temperature differs from 98.6°F?

Example 14.9: State and Plan

Let μ represent the population mean oral temperature of healthy adults in the population of interest.

STATE

The question concerns a difference in either direction, so the hypotheses are

$$H_0 : \mu = 98.6^\circ\text{F}$$

$$H_a : \mu \neq 98.6^\circ\text{F}$$

The significance level is $\alpha = 0.05$.

PLAN

A one-sample z test for a population mean will be used. The observations are assumed to be independent and representative. The sample size is large and the population standard deviation is treated as known.

Example 14.9: Standard Error and Test Statistic

The standard error is

$$\text{SE}(\bar{x}) = \frac{\sigma}{\sqrt{n}} = \frac{0.6}{\sqrt{130}} \approx 0.0526^\circ\text{F}$$

The observed difference from the null mean is

$$\bar{x} - \mu_0 = 98.25 - 98.6 = -0.35^\circ\text{F}$$

The test statistic is

$$z_{\text{obs}} = \frac{98.25 - 98.6}{0.6/\sqrt{130}} \approx -6.65$$

The observed sample mean is approximately 6.65 standard errors below the value specified by H_0.

Example 14.9: P-Value and Conclusion

Because the alternative hypothesis is two-sided, the p-value includes extreme results in both tails

$$P = 2P(Z \leq -6.65) \approx 2.9 \times 10^{-11}$$

The p-value is much smaller than the significance level

$$2.9 \times 10^{-11} < 0.05$$

Therefore, we reject H_0.

Conclusion

The data provide extremely strong evidence that the population mean oral temperature of healthy adults in the population of interest differs from 98.6°F. The sample result suggests that the population mean is lower than the historical reference value.

Example 14.9: Body Temperature Test in R

```
# Null value and known population standard deviation
mu_0 <- 98.6
sigma <- 0.6

# Sample information
x_bar <- 98.25
n <- 130

# Standard error and test statistic
standard_error <- sigma / sqrt(n)
z_observed <- (x_bar - mu_0) / standard_error

# Two-sided P-value
p_value <- 2 * pnorm(-abs(z_observed))

# Display the results
results <- c(standard_error = standard_error, z_statistic = z_observed, p_value = p_value)
signif(results, 4)
standard_error     z_statistic          p_value
      5.262e-02         -6.651        2.900e-11
```

The Confidence Interval and Test Connection

Two-sided confidence intervals and two-sided hypothesis tests are closely connected when they use the same statistical model and standard error.

A two-sided hypothesis test with significance level α corresponds to a confidence interval with confidence level

$$C = 1 - \alpha$$

Significance level α	**Confidence level** C
0.10	90%
0.05	95%
0.01	99%

The confidence interval identifies the parameter values that remain reasonably compatible with the sample data. The hypothesis test evaluates one particular value from that set of possibilities.

Using a Confidence Interval to Test a Null Value

Consider a two-sided hypothesis test

$$H_0 : \mu = \mu_0$$

using significance level α. The corresponding $(1 - \alpha)100\%$ confidence interval leads to the same decision:

Location of μ_0	**Test decision**	**P-value**
Inside the interval	Fail to reject H_0	$P > \alpha$
Outside the interval	Reject H_0	$P < \alpha$

Equivalent Interpretation

The confidence interval contains the parameter values that would not be rejected by the corresponding two-sided hypothesis test.

Example 14.10: Body Temperature Confidence Interval

For the body temperature study, the sample information is

$$\bar{x} = 98.25^\circ\text{F} \quad n = 130 \quad \sigma = 0.6^\circ\text{F}$$

Question

What is the 95% confidence interval for the population mean oral temperature of healthy adults in the population of interest? How does this interval lead to the same conclusion as the two-sided test

$$H_0 : \mu = 98.6^\circ\text{F}$$

at the $\alpha = 0.05$ significance level?

Example 14.10: Calculating the Interval

For a 95% confidence interval, the Standard Normal critical value is

$$z^* = 1.960$$

The standard error is

$$\text{SE}(\bar{x}) = \frac{\sigma}{\sqrt{n}} = \frac{0.6}{\sqrt{130}} \approx 0.0526^\circ\text{F}$$

The margin of error is

$$m = z^* \, \text{SE}(\bar{x})$$

$$m = 1.960(0.0526) \approx 0.1031^\circ\text{F}$$

Therefore, the interval extends approximately 0.1031°F on each side of the sample mean.

Example 14.10: Interval and Test Conclusion

The 95% confidence interval is

$$\bar{x} \pm m$$

$$98.25 \pm 0.1031$$

Therefore

$$\boxed{(98.15,\ 98.35)^\circ\text{F}}$$

Confidence-Interval Interpretation

We are 95% confident that the population mean oral temperature of healthy adults in the population of interest is between approximately 98.15°F and 98.35°F.

The null value is outside the interval because

$$98.6 > 98.35$$

Therefore, the corresponding two-sided test rejects H_0 at $\alpha = 0.05$. Both procedures provide evidence that the population mean differs from 98.6°F.

Example 14.10: Confidence Interval in R

```
# Sample information
x_bar <- 98.25
n <- 130

# Known population standard deviation
sigma <- 0.6

# Standard error and 95% critical value
standard_error <- sigma / sqrt(n)
z_star <- qnorm(0.975)

# Margin of error
margin_error <- z_star * standard_error

# Confidence interval
confidence_interval <- c(lower = x_bar - margin_error, upper = x_bar + margin_error)
round(confidence_interval, 2)

lower upper
98.15 98.35
```